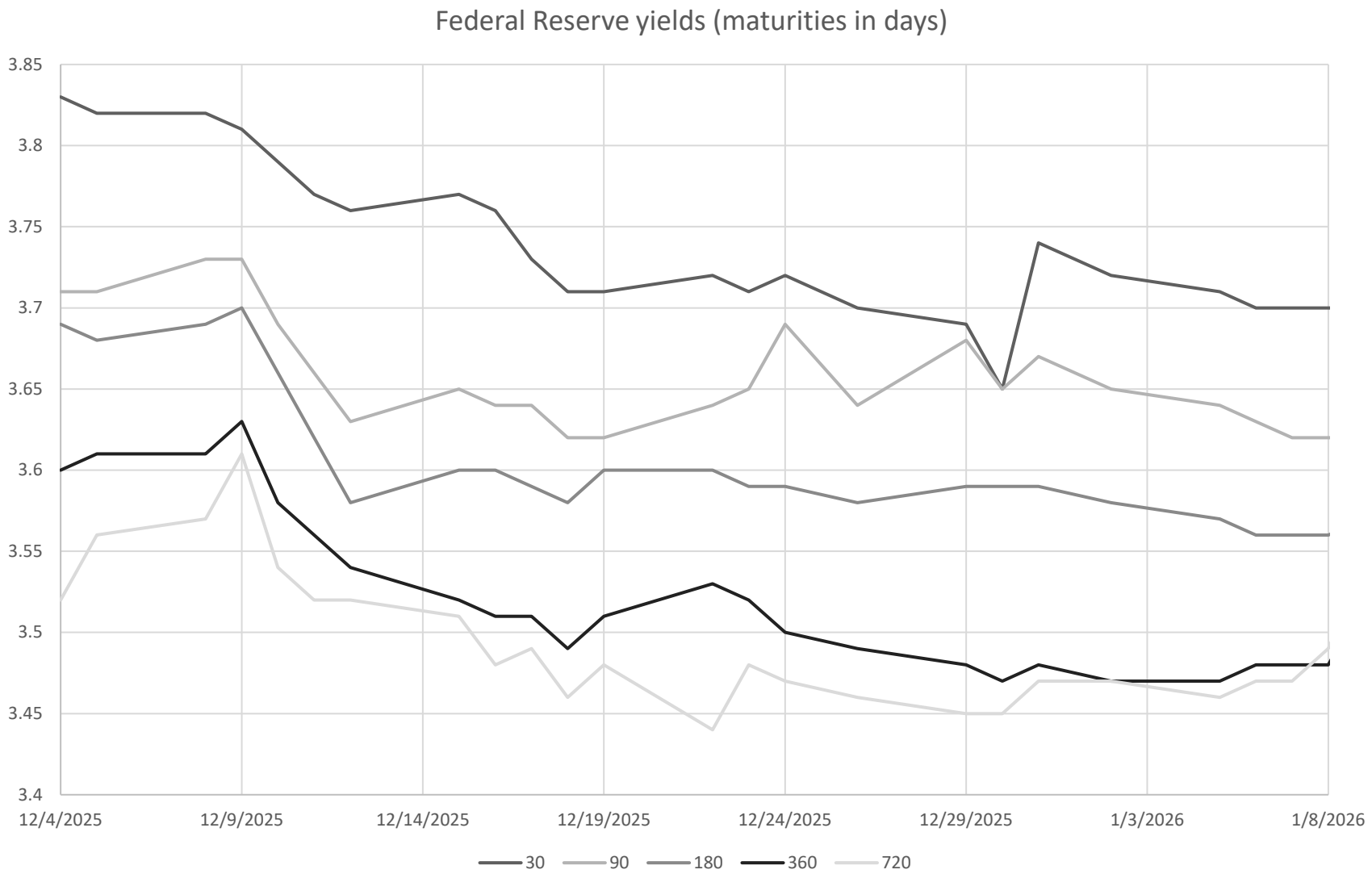
Real Finance



Class 04: Pricing Riskless Bonds

- Net Present value, again;
- Fed policy and inflation;
- Expectations and the yield curve;
- Implication for assets with risk;
- Application: QE2 and the unwinding of the Fed Position

Deciding value

Discounted present value

Core concept

Notice

(1) if different people (places) have different r, they have different discounted present values

(2) the same project presented in a period when interest rates are low is more attractive than when interest rates are high. (Value vs Growth investing)

(3) this is one reason central banks matter (they play an important role in setting r).

- Interest rate and discount rate $d = \frac{1}{1+r}$

$V = \sum_1^T \frac{x_t}{(1+r)^t} \quad = \sum_1^T x_t \, d^t$

Computing yield

- Interest rate is r per year. The price of a one year bond with no coupon is X =B/(1+r).
  - Now suppose you observe price (X)
  - Then yield price of a one year bond with no coupon is the interest rate such that r=(B/X)-1
  - That is the nominal yield
- General bond (face value B, coupon c, matures at T).
  - The yield (r) is the interest rate that solves
  - $X = \sum_{t=1}^{T} \frac{c}{(1+r)^t} + \frac{B}{(1+r)^T}$
- If you know X you can find r. if you know r you can find X

No arbitrage condition

NPV and Additivity

- A bond is a promise of 1 or more payments over time face value 100
- 5 year 3% coupon

Jan-26	Jan-27	Jan-28	Jan-29	Jan-30
3	3	3	3	103

- Additivity says the value of the bond has to be the sum of the value of cash flows

V=P	interest rate 2%				
$104.71	$2.94	$2.88	$2.83	$2.77	$93.29
	interest rate 4%				
$95.55	$2.88	$2.77	$2.67	$2.56	$84.66

- If markets are competitive price will be value
- What about if not?
  - Which side has more market power

Riskless bonds

- Definition
  - A bond that has a zero-default probability

Jan-26	Jan-27	Jan-28	Jan-29	Jan-30
3	3	3	3	103

  - Is this really riskless from the buyer's perspective?
- Not quite
  - Price level risk (inflation)
  - Price of time risk (Interest rate risk)
- Longer maturity => more risk
- This has implication that there is a yield curve
  - Get more per unit time for longer commitment

The yield cuve

- Two issues

1. interest rates over time with same maturity.

   In a simple model this is constant

   Anchored by the social discount rate (supply)

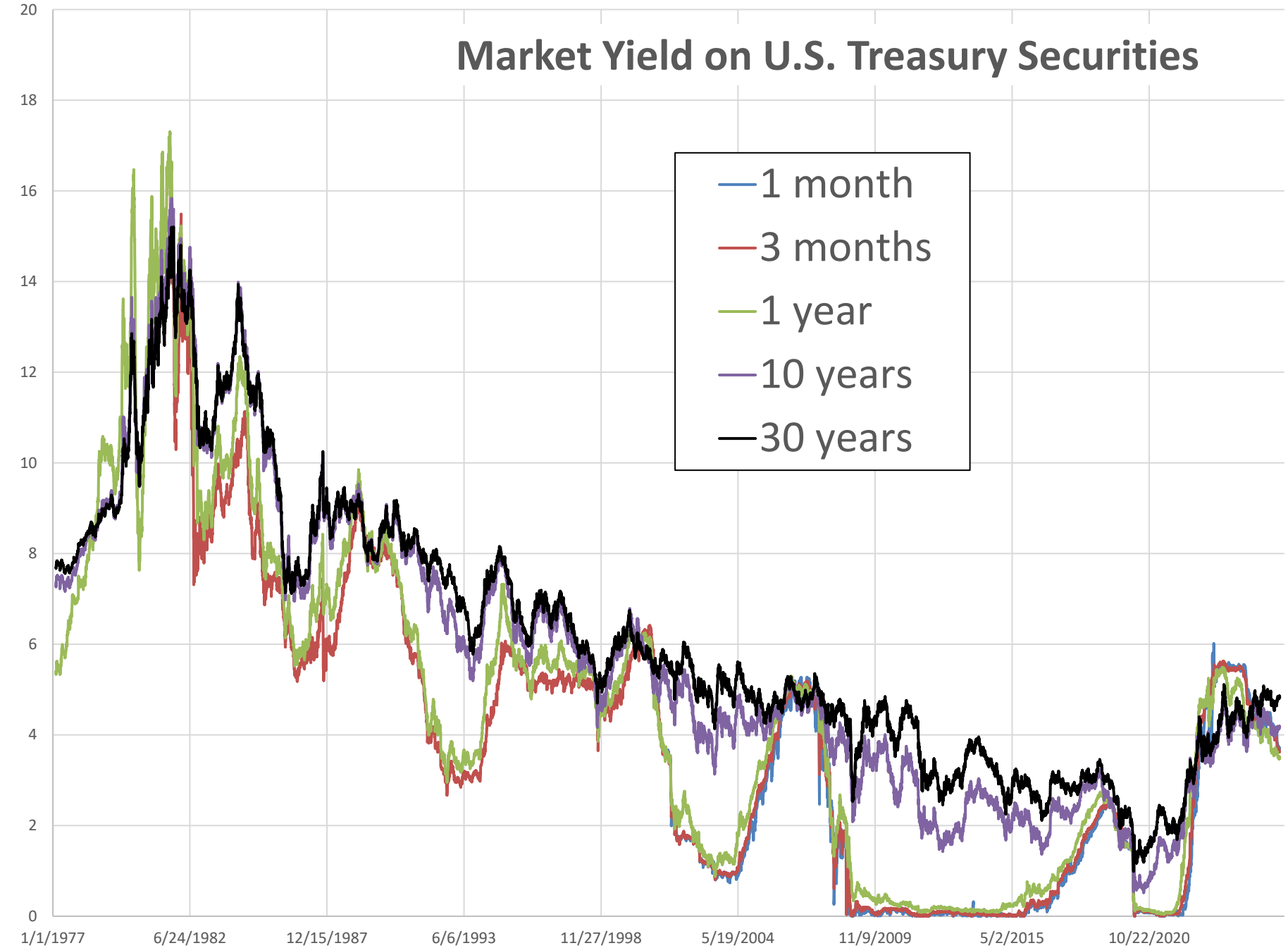
   Could be some cyclicality due to demand for capital

2. Interest rates at same time with different maturities

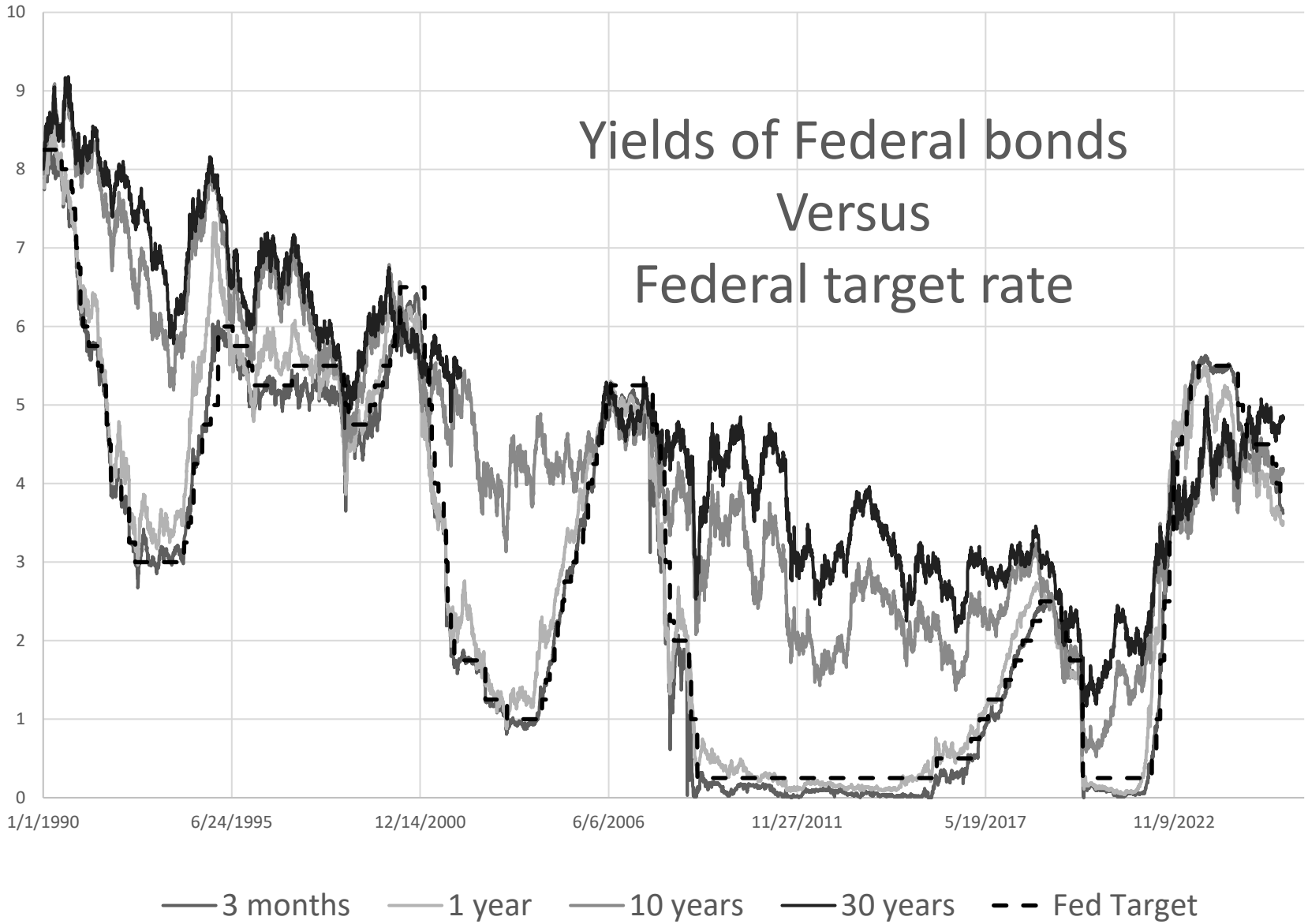
   1. Risk and liquidity

3. Federal government bonds

   1. The largest type of such bonds.

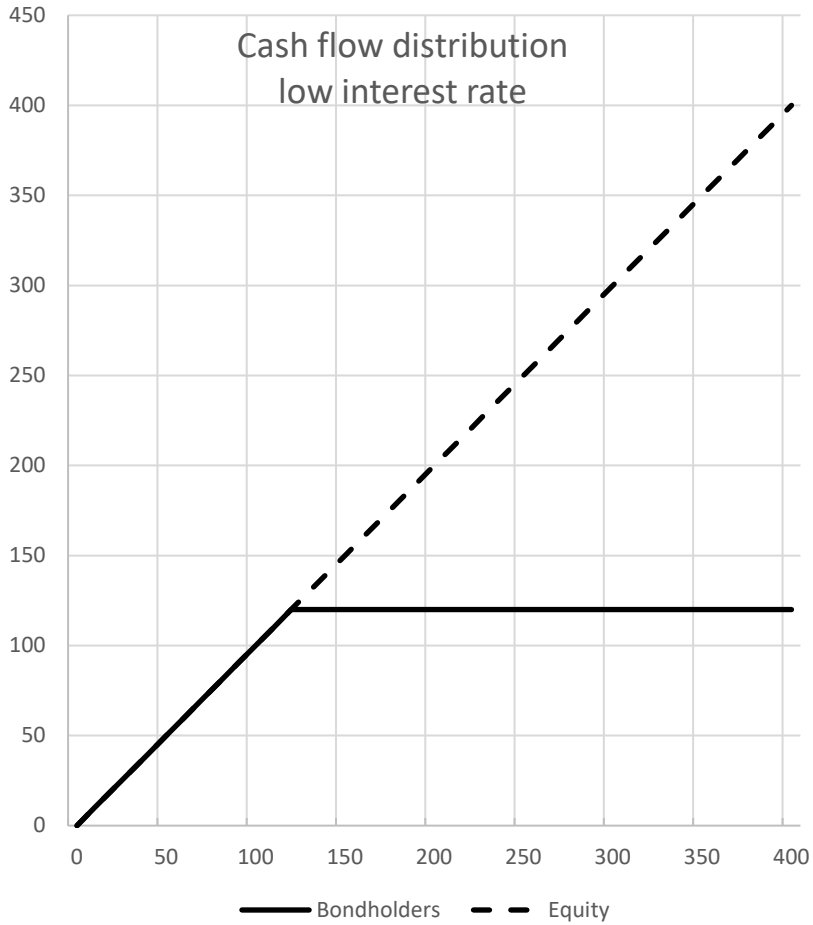


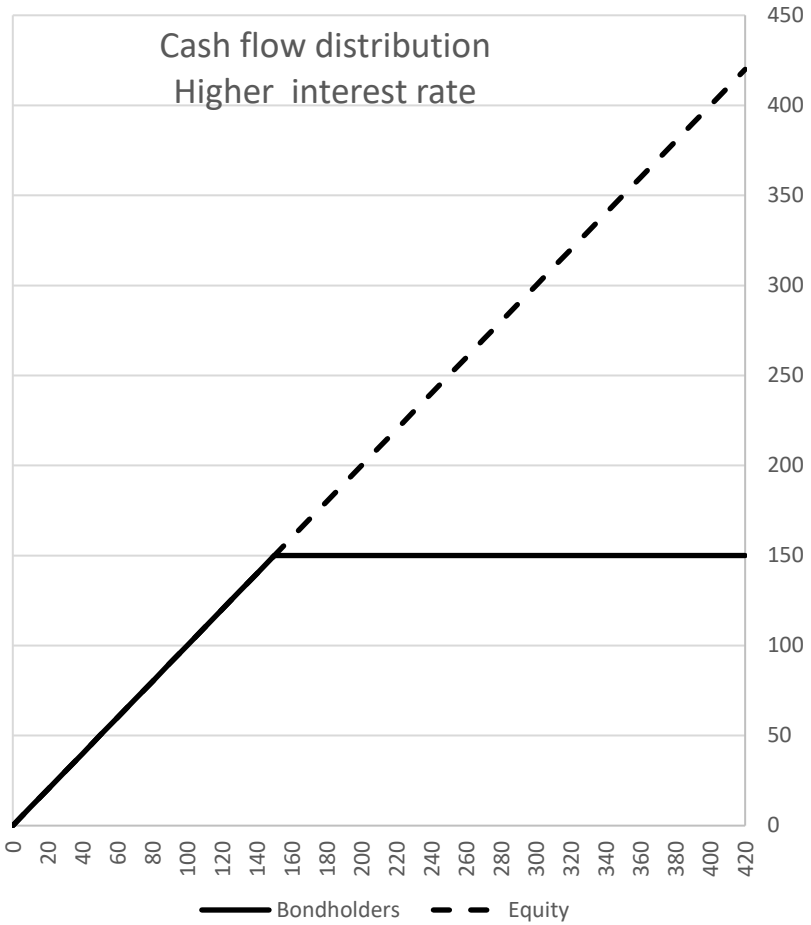
https://home.treasury.gov/resource-center/data-chart-center/interest-rates/

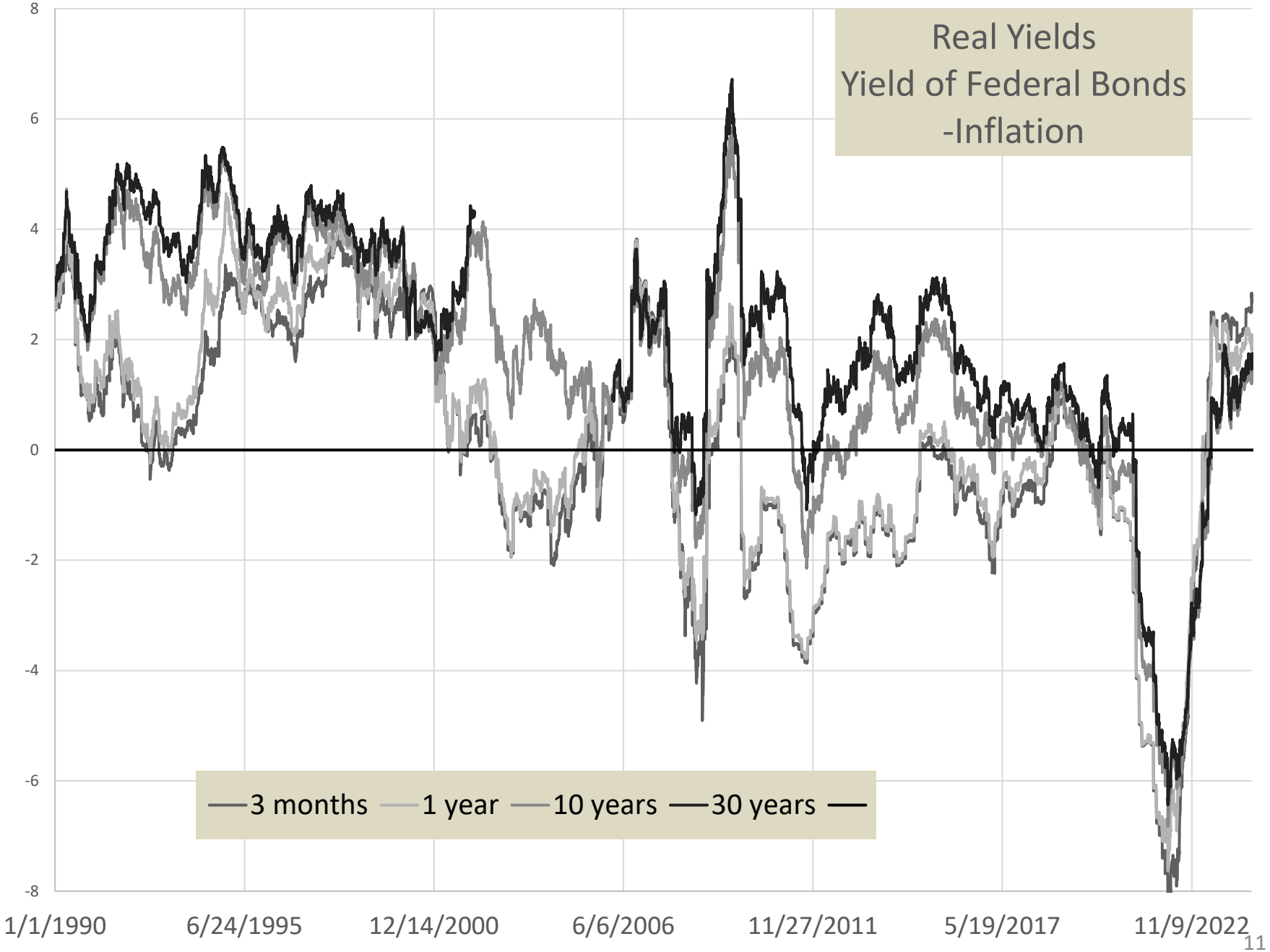


Short-term rates track Fed-target rate long-term rates do their own thing. Why

Debt and Equity



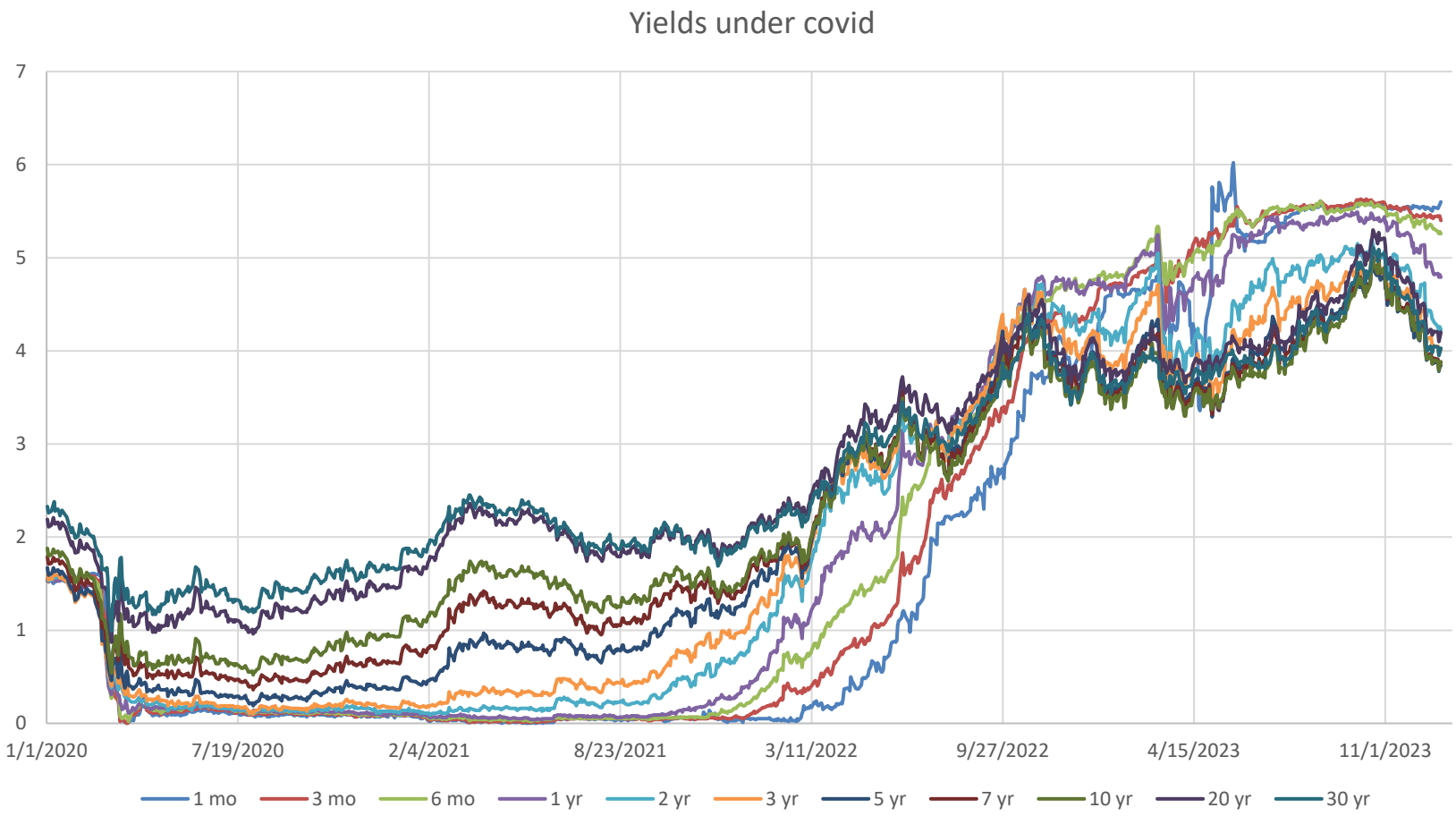



Real Yields
Yield of Federal Bonds
-Inflation
8
6
4
2
0
-2
-4
-6
-8
3 months
1 year
10 years
30 years
1/1/1990
6/24/1995
12/14/2000
6/6/2006
11/27/2011
5/19/2017
11/9/2022

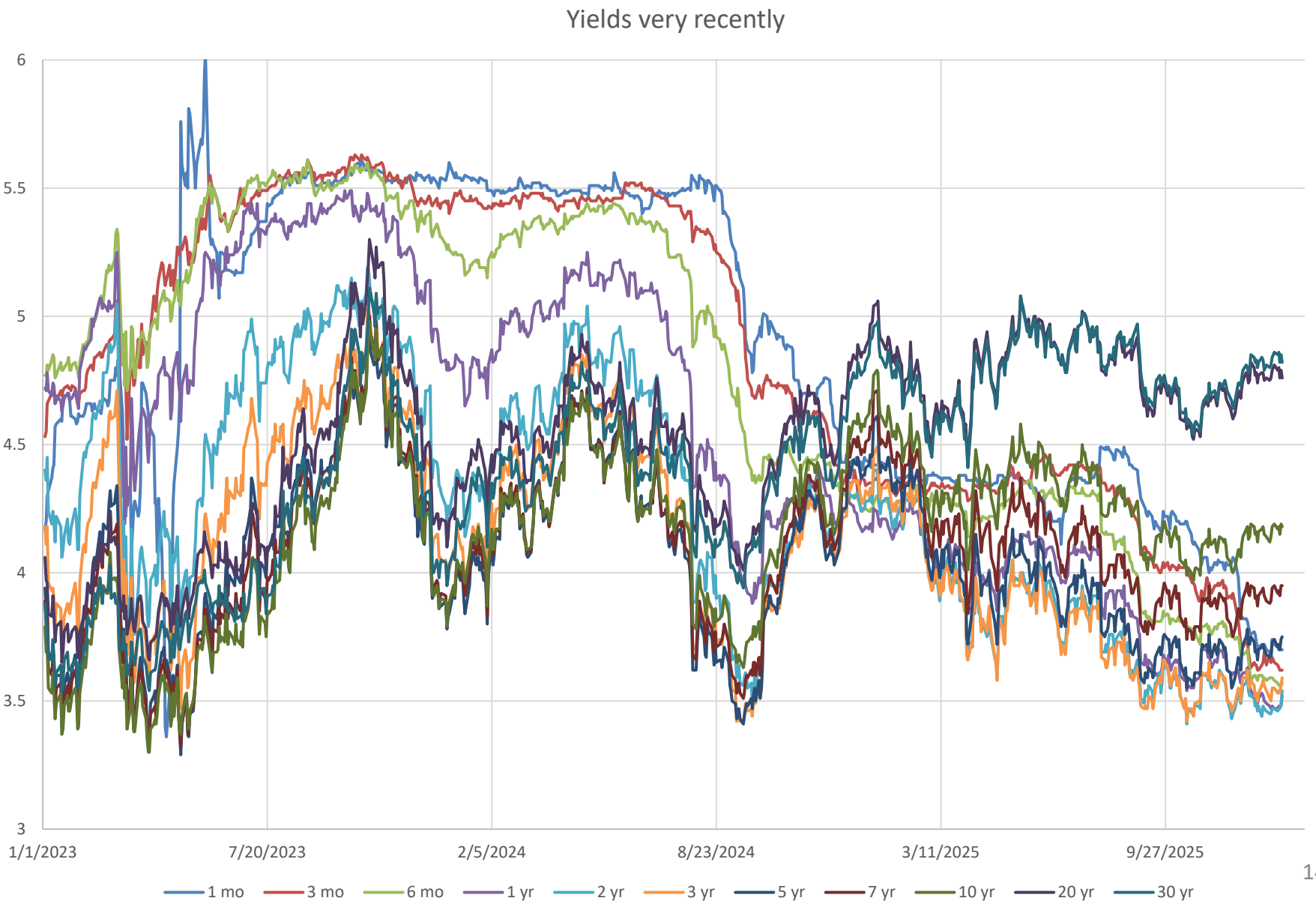
Things to remember

- For a ‘long’ time yields have been falling
- There are periods where the long yield (3 years or more) is much higher than the short yield (3 months or less)
  - There is a positive yield curve
- There are period where the long yield is very close to the short yield.
  - This happens both when short yields are high (1990) or lower (2006)
  - But not when short yields are very low(2001-2005, post 2008)
- Why?

From lock down to inflation



2023-2025


Yields very recently
6
5.5
5
4.5
4
3.5
3
1/1/2023
7/20/2023
2/5/2024
8/23/2024
3/11/2025
9/27/2025
1 mo
3 mo
6 mo
1 yr
2 yr
3 yr
5 yr
7 yr
10 yr
20 yr
30 yr

Thinks to Remember

- Long term decline in rates is one of the most important facts about recent finance
- The last years have seen a reversal
- Driven by inflation and monetary policy
  - Massive injection of liquidity in system during covid (quantitative easing)
  - Federal reserve raises rates

Explaining these patterns

- Two steps
- Suppose these bonds are riskless, how should they be priced?
  - Problem for today

- Then bring uncertainty and default risk back in.

  Problem for next class

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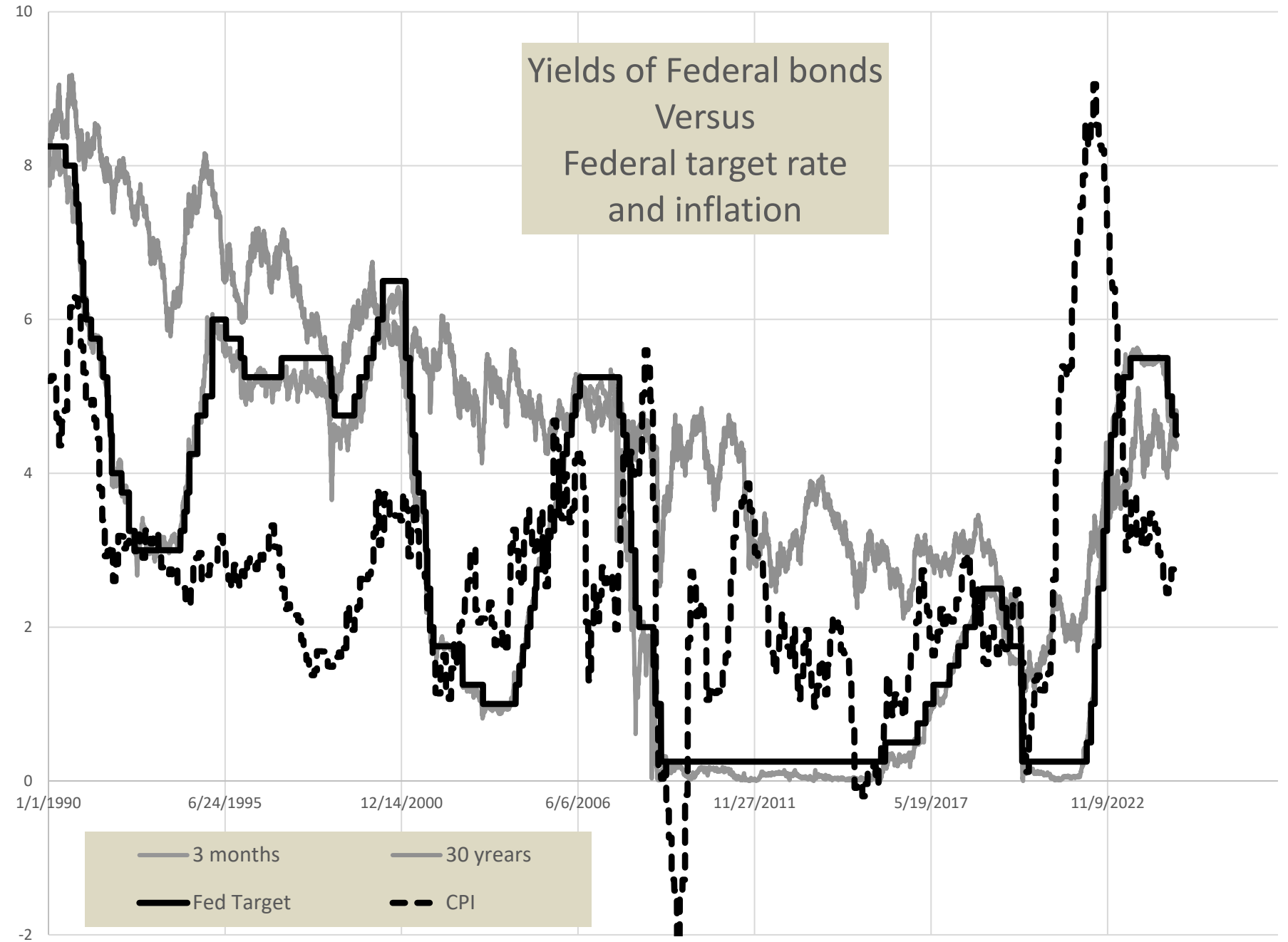
Definitions

- Riskless bond: A bond that has a zero probability of not making payments on time.
- What about inflation?
- Every other source of uncertainty creates risky bonds


Yields of Federal bonds
Versus
Federal target rate
and inflation
10
8
6
4
2
0
-2
1/1/1990
6/24/1995
12/14/2000
6/6/2006
11/27/2011
5/19/2017
11/9/2022
3 months
30 yrears
Fed Target
CPI

Lets price a simple bond

- Government issues a 1 month treasury bill (no coupon)
- Pay B at time 1 (November 16). What is its price (x) at time 0 (October 16)?
- Investor choice
  - Hold bond or hold cash?
    - Price of bond is just B because the return on the alternative is zero.
  - Hold bond and consume B at time 1 or consume X at time 1.
    - Investor is impatient so discounts the future by d ($d<1$) So investor has to be paid to hold the bond ($X<B$)
    - Hold bond if $U(X)<d*U(B)$.
    - At margin indifferent. So $U(X)= d*U(B)$
    - Simplify: U is linear. Then $u(X)=X=dB$ or $X/B=d$.
  - But we do not consume money we consume goods. Bond pays in money so need to adjust for prices.
    - Let price level at time 0 be 100, and the price level at time 1 be p.
    - So $U(X)= d*U(B)$ becomes $U(X/100)= d*U(B/p)$
    - $X/B=d$ now becomes $X/B =d(100/p)$.
- If there is inflation $100/p<1$ so $d(100/p)<d$ investor must be paid to hold the bond for his value of time *and* inflation.

The yield curve without uncertainty

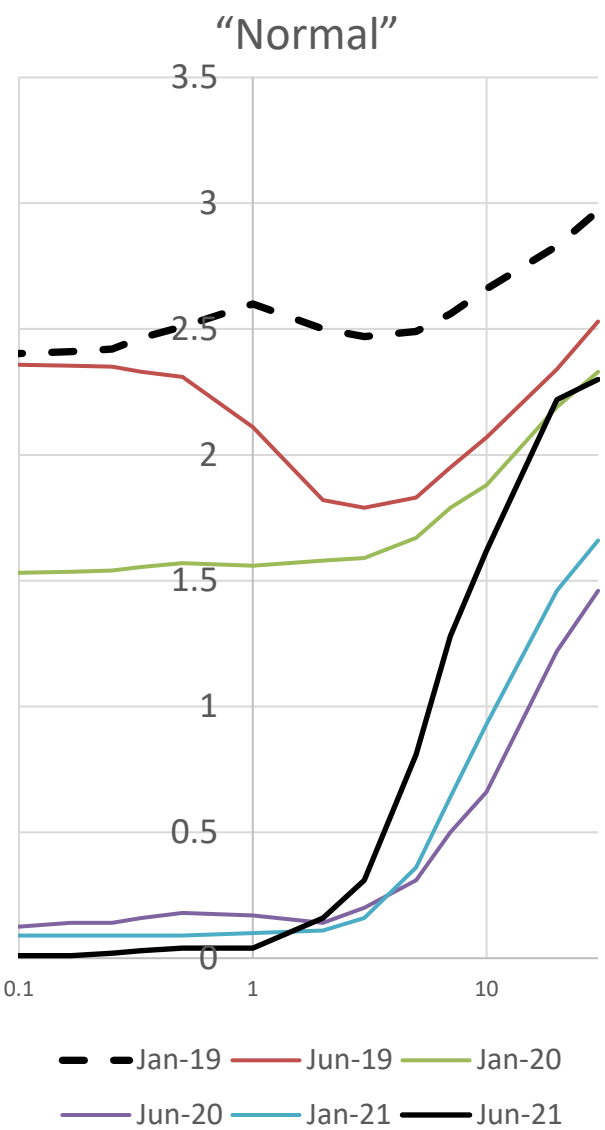
- Suppose the long-term interest rate is r per year and the change in the price level per year is ΔP. Then the price X of a one-year bond B with no coupon is X =B/(1+r).
- With coupon? one-year riskless bond that pays r X= $B\frac{1+r}{1+r} = B$
- Now suppose I offer a two-year riskless bond that pays r per year
- Its price is X= $B\frac{r}{1+r} + B\frac{(1+r)}{(1+r)^2}$ =
- $= B(\frac{r}{1+r} + \frac{1}{1+r}) = B(\frac{1+r}{1+r}) = B$

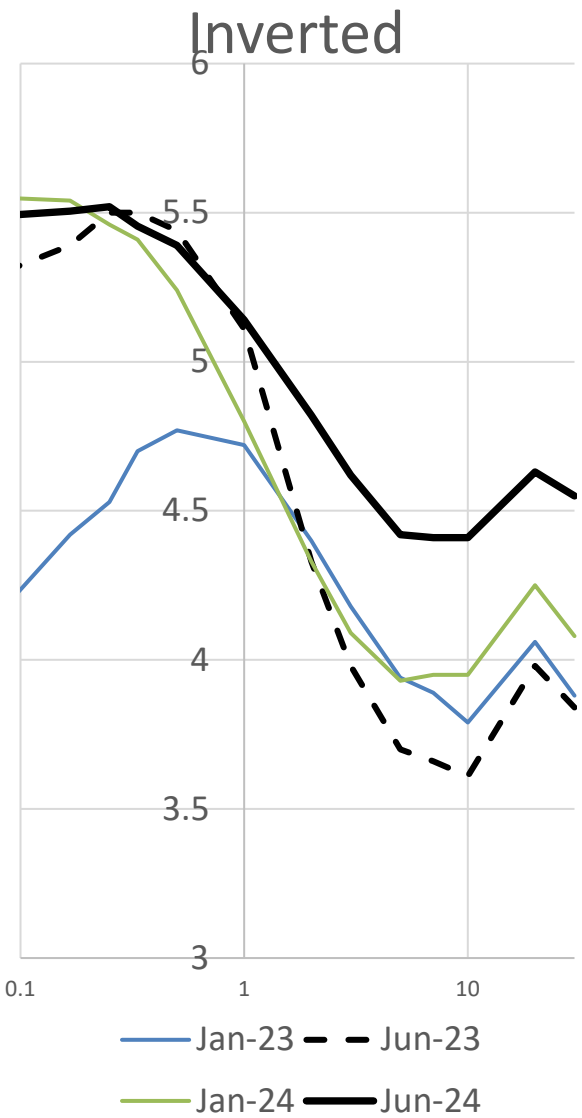
- Repeat as many periods as needed. You will see that the price of a bond riskless that pays r is constant. Its yield is r no matter what length the bond runs.
- If coupon pays less than r, the bond's price will fall with maturity but yield will be independent of maturity.
  - So why is there a yield curve

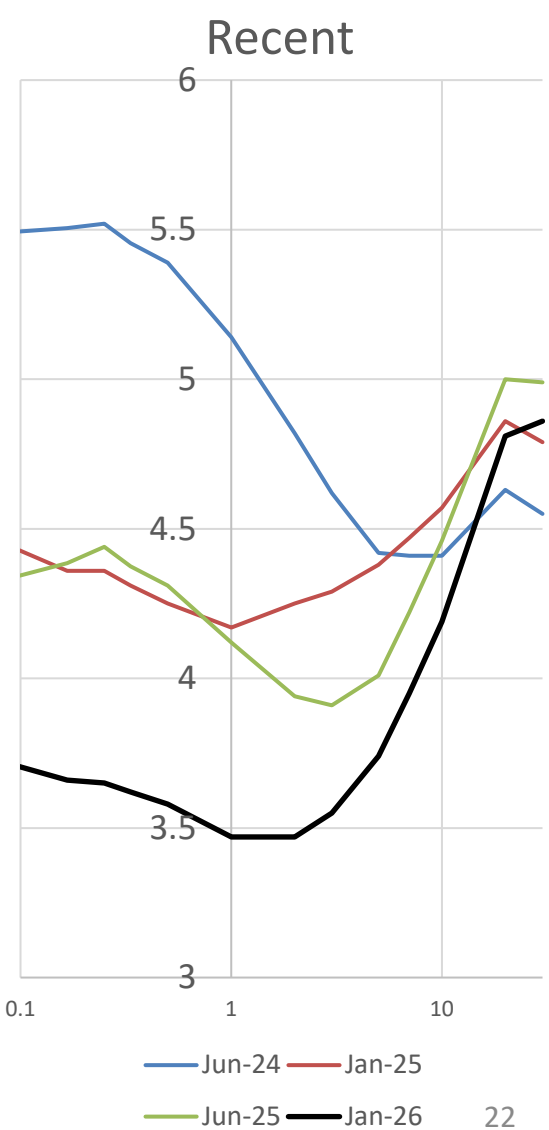
So much for Theory

- Prices are set to equalize yields both across types of bonds and over time.
- So absent something that is not evenly distributed there should be no yield curve
- So why do we see a yield curve about half the time in the data?
  - Federal reserve policy
  - Mean reversion in interest rates
  - The supply of funds is not infinitely elastic

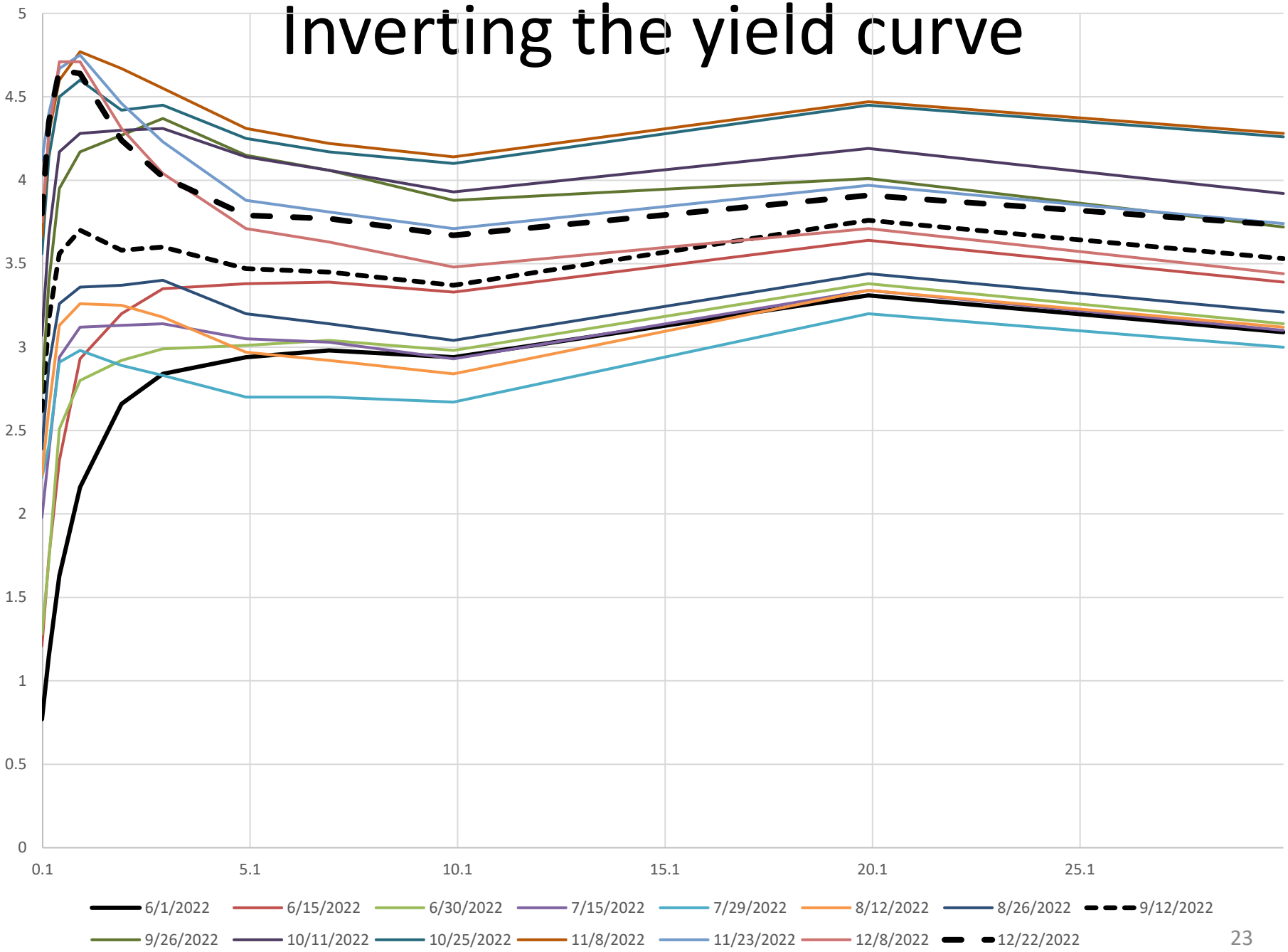
Yield curves







Inverting the yield curve



Implication for the Yield curve

- Suppose the Federal Reserve is happiest when the economy grows between 2.5 and 3.5% and unemployment is between 3 and 5%.
- And the long-term interest rate is 4%
- These rates keeps inflation in control (and thus allows it to set interest rates)
- When there is a negative shock (growth goes down and unemployment goes up). The Fed lowers interest rates, and the short-term rate falls
- But the market knows that when the shock has passed the Federal Reserve will raise interest rates (so long rates not so sensitive).
- Notice now future interest rates are no longer symmetrically distributed around the current level, they have mean reversion around some long-term trend.

The yield curve and project selection

- Long-term rates => cost of Long-term investment,
  - (infrastructure, research)
  - Climate adaptation
- Other rates tell you when to do things
- Maintenance
- When you have inversion might want to go long
  - But what is the market telling you.
  - Can you call the bonds

Next time

- **<u>Class 5: Financial markets</u>**
  - **Key question where do prices come from**
  - **Define liquidity**
  - **Connect to competition**